Project 2: Analyzing Pittsburgh Police Incidents with Tableau

In this project, you will use Tableau to create interactive visualizations based on public data from the Western Pennsylvania Regional Data Center (WPRDC) related to Pittsburgh Police Incidents. This project will allow you to explore Tableau's capabilities for visual storytelling while providing insights into crime trends in Pittsburgh.

Project 2 is an extension of Lab 2: Tableau and Integration in Observable Dashboard. You will use the same datasets of Pittsburgh police incidents (blotter and arrests) to tell the story of Pittsburgh within the datasets. You must create two additional visualizations that differ from the visualizations we created in the Lab 2.

1. Objectives

- To practice using Tableau to analyze and visualize real-world public safety data.
- To explore trends or patterns in Pittsburgh crime data, focusing on effectively communicating insights.
- To develop interactive visualizations that engage viewers and highlight important aspects of the data.

2. Dataset

Access the Pittsburgh Police Incident Blotter Data and Arrest Data from Lab 2 (in the Lab 2 data folder). Their pages on the Western Pennsylvania Regional Data Center (WPRDC) explain the dataset variables and time ranges.

3. Defining Your Data Story

- Decide on a specific focus or "story" within the dataset, such as crime trends over time, hotspots by neighborhood, seasonal or hourly patterns, or types of incidents.
- Ensure your story is supported by data and that it provides meaningful insights into public safety in Pittsburgh.

4. Data Analysis and Dashboard Design

Based on the in-class lab on Tableau, develop two more ***additional*** interactive visualizations that convey different perspectives or aspects of the data. For example (you can choose your own visualizations):

- A time-series chart displaying trends over time with filters for incident type or location.
- A map visualization showing incident hotspots or patterns by neighborhood.

Your visualizations should be interactive, featuring filters, tooltips, or other options that enable viewers to explore the data in greater detail.

5. Visualization consideration

- Focus on telling a story, not just displaying charts.
- Use consistent color schemes and clearly labeled elements.
- Keep interactivity simple and intuitive
- Cite all data sources

6. Storytelling and Reflection

Write a short (300 – 500 word) reflection of your design:

- Your approach to identifying a data story and the insight gained
- The rationale behind your design and interaction choices
- Challenges you encountered and how you addressed them.

7. Submission

Submit on Canvas a Word document containing:

- A link to your Tableau Public dashboard
- Your storytelling and reflection

8. Grading Rubric

Category	Criteria	Points
Data Story Definition	Relevance and clarity of the chosen data story, supported by specific trends or patterns.	3
Visualization Quality	Effectiveness of visual encoding, chart types, and clarity in each visualization to convey key insights.	4
Interactivity	Use of interactive features (e.g., filters, tooltips) to allow viewers to explore different aspects.	4
Design and Storytelling	Application of design principles and logical narrative structure across both visualizations.	4
Technical Accuracy	Data accuracy, appropriate use of Tableau tools, and correct configuration of interactive elements.	3
Reflection	Depth of insight in discussing design decisions, challenges, and key takeaways in the reflection.	2